

Agent Memory Integrity Standard - (AMIS)

OOF™ Origin Open Foundation™

Independent Methodological Authority

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Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Memory Governance Intelligence Architecture (MGIA™)

Operational Layer: Agent Memory Governance Layer

Governed Space: Agent Memory Integrity

Category: AI & Interpretation

Subcategory: Agent Memory Governance Architecture

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Compatibility: OOF Methodology OS · Memory Integrity Standard (MIS) ·

Agent Cognition Integrity Standard (ACIS) ·

Human-AI Cognition Integrity Standard (HAICS) ·

Cognitive Evolution Governance Standard (CEGS) ·

INTEGROS® — Integrity Standard

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition System

Canonical Definition

Agent Memory Integrity Standard (AMIS) defines the structural conditions under which memory utilized by autonomous agents remains continuous, retrievable, updateable, coherent, accountable, reality-connected, governable, and operationally valid throughout agent operation, learning, adaptation, collaboration, and decision-support activities.

AMIS governs agent memory.

The standard establishes the integrity conditions required for autonomous agents to preserve operational continuity, contextual awareness, historical understanding, learned knowledge, and memory-based reasoning across time.

Agent memory is not merely storage.

Agent memory is the operational persistence layer supporting autonomous cognition.

AMIS governs the integrity of that persistence.

A. Standard Abstract

Autonomous agents depend on memory.

Without memory an agent cannot:

- maintain context
- preserve experience
- learn effectively
- sustain goals
- reason across time
- coordinate complex tasks
- maintain operational continuity

As AI agents become increasingly autonomous, memory becomes one of the most critical operational layers.

This creates a governance challenge.

How can agent memory remain reliable?

How can agents update memory without corruption?

How can memory remain useful over long periods?

How can agent memory remain accountable?

AMIS exists to govern these conditions.

B. Core Principle

Agent memory becomes valid only when memory continuity, retrieval, updating, consistency, accountability, and operational utilization remain traceable, governable, reality-connected, and operationally valid throughout agent operation.

C. Scope

This standard may apply to:

- autonomous agents
- AI assistants
- agent ecosystems
- enterprise agents
- robotics agents
- multi-agent environments
- Human-AI agents
- future autonomous intelligence systems

AMIS applies wherever agents utilize memory to support

cognition and operation.

D. Why This Standard Exists

Modern agents increasingly rely on memory.

Memory enables agents to:

- preserve context
- maintain continuity
- accumulate experience
- adapt over time
- coordinate actions
- support reasoning

Without governance:

memory corruption may occur

context may be lost

retrieval may fail

updates may become unreliable

agent behavior may drift

AMIS exists because agent memory itself has become a governable space.

E. Agent Memory Integrity Logic

Agent memory integrity exists only when the following remain materially preservable:

- **1. Agent Memory Continuity Integrity:**
Memory remains available across operational time.
 - **2. Agent Memory Retrieval Integrity:**
Memory remains retrievable when needed.
 - **3. Agent Memory Update Integrity:**
Memory modifications remain governable.
 - **4. Agent Memory Consistency Integrity:**
Memory remains coherent and non-contradictory.
 - **5. Agent Memory Accountability Integrity:**
Memory utilization remains attributable and reviewable.
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F. Operational Architecture Space

AMIS defines the operational architecture space for:

- agent memory governance
- memory continuity
- memory retrieval
- memory updates
- memory consistency
- memory accountability
- contextual persistence
- agent memory utilization

This space exists because autonomous agents depend on memory persistence to remain operationally effective.

AMIS governs whether that persistence remains valid.

G. Difference Between Agent Cognition and Agent Memory

Agent cognition governs how an agent thinks.

Agent memory governs what persists between thinking events.

- ACIS governs cognition integrity.
- AMIS governs memory integrity.

Both are required for trustworthy autonomous agents.

H. Runtime Position

AMIS operates beneath agent cognition.

It provides the memory persistence layer required by:

- reasoning
- planning
- interpretation
- decision support
- adaptation
- learning

AMIS therefore serves as the memory foundation of autonomous agent operation.

I. Agent Memory Failure Rule

Agent memory failure occurs when memory can no longer reliably support cognition.

Examples include:

- context loss
- retrieval failure
- contradictory memory
- corrupted memory
- memory drift
- uncontrolled updates

AMIS exists to expose and govern these conditions.

J. Validity Logic

An agent memory environment is valid under AMIS when:

- memory remains continuous
- memory remains retrievable
- updates remain governable
- memory remains coherent
- memory remains accountable
- memory remains operationally useful

An agent memory environment becomes invalid when:

- context disappears
- retrieval becomes unreliable
- updates become uncontrolled
- contradictions accumulate
- accountability is lost

K. Relationship to Other OOF Standards

AMIS derives from:

Memory Integrity Standard (MIS)

and directly supports:

- Agent Cognition Integrity Standard (ACIS)
- Multi-Agent Cognition Integrity Standard (MCIS)
- Human-AI Cognition Integrity Standard (HAICS)
- Cognitive Evolution Governance Standard (CEGS)

AMIS governs memory persistence.

ACIS governs cognition built upon that persistence.

L. Foundational Principle

An autonomous agent may continue operating after memory integrity is lost, but its cognition progressively loses continuity, context, and reliability.

Canonical Closing Statement

Agent Memory Integrity Standard (AMIS) defines the structural conditions under which memory utilized by autonomous agents remains continuous, retrievable, updateable, coherent, accountable, reality-connected, governable, and operationally valid throughout agent operation, learning, adaptation, collaboration, and decision-support activities.

Autonomous cognition depends on memory persistence. Agent memory integrity therefore becomes the foundational memory condition of trustworthy autonomous agents.

Module Architecture

[→ AMAM — Agent Memory Accountability Module](#)

[→ AMCM — Agent Memory Continuity Module](#)

[→ AMCOM — Agent Memory Consistency Module](#)

[→ AMRM — Agent Memory Retrieval Module](#)

[→ AMUM — Agent Memory Update Module](#)

Related Documents

[→ About the Agent Memory Integrity Standard \(AMIS\).](#)

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